Exercise 1: Database searches (8p)

a) **Retrieve the protein sequence with accession number NP_081220.1 from the National Center for Biotechnology Information (NCBI) using their website at https://www.ncbi.nlm.nih.gov/ and its search functions. The sequence is from mouse (Mus musculus) and should be 286 amino acids long. Save it in a FASTA-formatted text file. Show the formatted sequence in the report. (1p)**

```
>NP_081220.1 alpha-ketoglutarate-dependent dioxygenase
alkB homolog 3 isoform a [Mus musculus]
MEDKRQRARVQGGWATPTKSQSATQPASPARSRLSQTAGPAWRSKEQQQCDRQFVFK
EPQLVVRAAPEPR
VIDREGVYEISLSPTGVSRVCLYPGFVDLKEADWILEQLCKDVPWKQRMGIREDVTY
PQPRLTAWYGELP
YTYSRITMEPNPHWLPVLWTLKSRIEENTSHTFNSLLCNFYRDEKDSVDWHSDDEPS
LGSCPVIASLSFG
ATRTFEMRKKPPPEENGDYTYVERVKIPLDHGTLLIMEGATQADWQHRVPKEYHSRQ
PRVNLTFRTVYPD
PRGAPR
```

b) **Which variant of BLAST should be used to search the UniProKB/Swiss-Prot database with the retrieved sequence as the query? (1p)**

BLAST P, because the query is a protein sequence containing aminoacids and the uniprokb/swiss-prot is a protein database.

c) **Use BLAST at https://blast.ncbi.nlm.nih.gov/Blast.cgi to search the UniProKB/Swiss-Prot database with the retrieved sequence as the query. Algorithm parameters must be adjusted as follows: "Expect threshold" = 10, "Word size" = 3, "Compositional adjustments" = "No adjustment" and Filter "Low complexity regions" = OFF. The scoring matrix should be BLOSUM62, and gap costs should be "Existence: 11 Extension: 1". How many hits do youget? (1p)**

Job Title	**Protein Sequence**
RID	UAX15TG4014 *Search expires on 03-03 18:27 pm* Download All
Program	BLASTP Citation
Database	swissprot See details
Query ID	lcl|Query_11669249
Description	unnamed protein product
Molecule type	amino acid
Query Length	286
Other reports	Distance tree of results Multiple alignment MSA viewer

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments

select all *17 sequences selected*

d) **What is the accession number (sequence id) of the best match with a sequence from Arabidopsis thaliana? Save the sequence in a file. (1p)**
Q9SIE0.2

e) **Show the BLAST alignment between the query protein and the Arabidopsis thaliana protein. (1p)**

RecName: Full=DNA oxidative demethylase ALKBH2; AltName: Full=Alkylated DNA repair protein alkB homolog 2; AltName: Full=Alpha-ketoglutarate-dependent dioxygenase alkB homolog 2 [Arabidopsis thaliana]

Sequence ID: Q9SIE0.2 Length: 314 Number of Matches: 1

Range 1: 113 to 313 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Identities	Positives	Gaps
92.4 bits(228)	2e-20	66/201(33%)	94/201(46%)	19/201(9%)

```
Query  96   FVDLKEADWILEQLCKDVPWKQRMGIREDVTYPQPRLTAWYGE-----LPYTYSRITMEP  150
            F+  +++    + L K +PW +          +  QPR T +        L Y+  R T
Sbjct  113  FLPFQQSWTFFDYLDKHIPWTRPTIRVFGRSCLQPRDTCYVASSGLTALVYSGYRPTSYS  172

Query  151  NPHWLPVLWTLKSRIEENTSHTFNSLLCNFYRDEKDSVDWHSDDEPSLGSCPVIASLSFG  210
               + P+   L +  +      FNSLL N Y+   D V WH+DDE   G  P IAS+SFG
Sbjct  173  WDDFPPLKEILDAIYKVLPGSRFNSLLLNRYKGASDYVAWHADDEKIYGPTPEIASVSFG  232

Query  211  ATRTFEMRKKPPPE----ENGD----------YTYVERVKIPLDHGTLLIMEGATQADWQ  256
              R F ++KK   E    + GD           +  ++  + L HG+LL+M G TQ DW
Sbjct  233  CERDFVLKKKKDEESSQGKTGDSGPAKKRLKRSSREDQQSLTLKHGSLLVMRGYTQRDWI  292

Query  257  HRVPKEYHSRQPRVNLTFRTV  277
            H VPK   +   R+NLTFR V
Sbjct  293  HSVPKRAKAEGTRINLTFRLV  313
```

Related Information
Gene - associated gene details
AlphaFold Structure - 3D structure displays

f) **What is the score (raw alignment score), the bit score and the E-value of this alignment? (3p)**
Score: 92.4 bits(228)
E-Value: 2e-20

Exercise 2:

a)

b) **Run your program on the mouse sequence and the Arabidopsis thaliana sequence. What score do you get? Show the alignment. (2p)**

```
Best local alignment score: 228
Seq1 region: 96..277
Seq2 region: 113..313
Alignment length: 201

Query     96  FVDLKEADWILEQLCKDVPWKQRMGIREDVTYPQPRLTAWYGE-----LPYTYSRITMEP  150
              |+  +++    + | | +|| +        +  ||| | +        | |+  | |
Sbjct    113  FLPFQQSWTFFDYLDKHIPWTRPTIRVFGRSCLQPRDTCYVASSGLTALVYSGYRPTSYS  172

Query    151  NPHWLPVLWTLKSRIEENTSHTFNSLLCNFYRDEKDSVDWHSDDEPSLGSCPVIASLSFG  210
                 + |+   | +  +      ||||| | |+   | | ||+|||   |  | |||+|||
Sbjct    173  WDDFPPLKEILDAIYKVLPGSRFNSLLLNRYKGASDYVAWHADDEKIYGPTPEIASVSFG  232

Query    211  ATRTFEMRKKPPPE----ENGD----------YTYVERVKIPLDHGTLLIMEGATQADWQ  256
                | | ++||   |    + ||           +  ++  + | ||+||+| | || ||
Sbjct    233  CERDFVLKKKKDEESSQGKTGDSGPAKKRLKRSSREDQQSLTLKHGSLLVMRGYTQRDWI  292

Query    257  HRVPKEYHSRQPRVNLTFRTV  277
              | |||   +   |+||||| |
Sbjct    293  HSVPKRAKAEGTRINLTFRLV  313
```

c) **Compare the score you got with the raw score from BLAST in exercise 1 and comment on any discrepancies. (2p)**
Score from exercise 1: 92.4 bits(228)
Score from program: 228

d) **Compare the actual alignment you got with the alignment from BLAST and comment on any discrepancies. (2p)**

```
Query     96  FVDLKEADWILEQLCKDVPWKQRMGIREDVTYPQPRLTAWYGE-----LPYTYSRITMEP  150
              |+  +++    + | | +|| +        +  ||| | +        | |+  | |
Sbjct    113  FLPFQQSWTFFDYLDKHIPWTRPTIRVFGRSCLQPRDTCYVASSGLTALVYSGYRPTSYS  172

Query    151  NPHWLPVLWTLKSRIEENTSHTFNSLLCNFYRDEKDSVDWHSDDEPSLGSCPVIASLSFG  210
                 + |+   | +  +      ||||| | |+   | | ||+|||   |  | |||+|||
Sbjct    173  WDDFPPLKEILDAIYKVLPGSRFNSLLLNRYKGASDYVAWHADDEKIYGPTPEIASVSFG  232

Query    211  ATRTFEMRKKPPPE----ENGD----------YTYVERVKIPLDHGTLLIMEGATQADWQ  256
                | | ++||   |    + ||           +  ++  + | ||+||+| | || ||
Sbjct    233  CERDFVLKKKKDEESSQGKTGDSGPAKKRLKRSSREDQQSLTLKHGSLLVMRGYTQRDWI  292

Query    257  HRVPKEYHSRQPRVNLTFRTV  277
              | |||   +   |+||||| |
Sbjct    293  HSVPKRAKAEGTRINLTFRLV  313
```

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```
Query  96   FVDLKEADWILEQLCKDVPWKQRMGIREDVTYPQPRLTAWYGE-----LPYTYSRITMEP  150
            F+  +++    + L K +PW +        +  QPR T +        L Y+  R T
Sbjct  113  FLPFQQSWTFFDYLDKHIPWTRPTIRVFGRSCLQPRDTCYVASSGLTALVYSGYRPTSYS  172

Query  151  NPHWLPVLWTLKSRIEENTSHTFNSLLCNFYRDEKDSVDWHSDDEPSLGSCPVIASLSFG  210
              + P+   L +  +      FNSLL N Y+   D V WH+DDE   G  P IAS+SFG
Sbjct  173  WDDFPPLKEILDAIYKVLPGSRFNSLLLNRYKGASDYVAWHADDEKIYGPTPEIASVSFG  232

Query  211  ATRTFEMRKKPPPE----ENGD----------YTYVERVKIPLDHGTLLIMEGATQADWQ  256
              R F ++KK   E    + GD           +  ++  + L HG+LL+M G TQ DW
Sbjct  233  CERDFVLKKKKDEESSQGKTGDSGPAKKRLKRSSREDQQSLTLKHGSLLVMRGYTQRDWI  292

Query  257  HRVPKEYHSRQPRVNLTFRTV  277
            H VPK   +   R+NLTFR V
Sbjct  293  HSVPKRAKAEGTRINLTFRLV  313
```

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The only descrepancy between the two alignments is that matching aminoacids, from my code, are marked as | instead of written out.

e) **Retrieve the ALKB_ECOLI (P05050) sequence from Escherichia coli in FASTA format. Use your program to compare this sequence with the mouse sequence from exercise 1a. What score do you get? Show the alignment. Why did the sequence from E. coli not show up in the search in exercise 1? (4p)**

```
Best local alignment score: 48
Seq1 region: 191..275
Seq2 region: 131..210
Alignment length: 98

Query    191  HSD-DEPSLGSCPVIASLSFGATRTFEMRKKPPPEENGDYTYVERVKIPLDHGTLLIMEG  249
              | | ||| | +  |  ||   |   |   |+  | +          ++ |+|| +++  |
Sbjct    131  HQDKDEPDLRAPIVSVSLGLPAIFQFGGLKRNDPLK---------RLLLEHGDVVVWGG  180

Query    250  ATQADWQHRVPKEYHSRQP-----------RVNLTFR  275
               ++          ||  ||            | |||||
Sbjct    181  ESRL--------FYHGIQPLKAGFHPLTIDCRYNLTFR  210
```

Delcartion of use of AI:
I have used Claude to help with programming the alghoritm and the blast like showcase.

Approach to excersie 1 was to follow along the lectures, doing each sub question as they were taught or went through in the lectures and group excersices.

Appraoch to excersice 2 was reading about the smith waterman algorithm in the book, looking at other github repos for inspiration, and combinding the information from both to do it mostly by myself, and “perfecting”/refining it with the use of Claude afterwards. The blast like print was done with mostly use of Claude, with some samll tweaks.